Time Is the Baseline to Beat: Calibrated Fusion of Policy-Internal Scores and Elapsed Time for VLA Failure Detection

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Abstract—Runtime failure detectors for vision-language-action (VLA) policies are rarely compared against elapsed time. Failed rollouts tend to run until the task timeout, so a task-conditioned time-risk detector that sees no policy features catches 63% of failures at a 5% false-positive rate (FPR) on 38 RoboCasa365 tasks with Xiaomi-Robotics-1, more than the SAFE failure predictor alone (57%). We propose time-aware fusion: alarm when either a policy-internal score or the time risk crosses its threshold, with both thresholds calibrated jointly on a separate cohort and evaluated once on a disjoint prospective cohort. [PENDING: one sentence: preregistered confirmatory result on 760 fresh rollouts.] On the prospective cohort used to develop the design, fusion raised recall to 0.955 at 0.058 FPR. Four success-only anomaly scores fused the same way reached 0.905–0.947, with no significant difference from fused SAFE, so the gain comes from the fusion rather than from a particular learned score. The gain is not an artifact of failed rollouts running longer: random noise fused with time under the same calibration recovers only the time-only recall (0.635 on average over 50 draws). Alarms remain late: no detector catches more than 3% of failures within the first quarter of the task horizon, and fused alarms fire a median of 12 policy inferences before the rollout ends.

I. Introduction

A failure detector earns its place only if it beats what a robot already knows for free: how long the current attempt has been running. Generalist VLA policies still fail often on household tasks, and a deployed system needs an online signal that a rollout is going wrong. Recent detectors read the policy's internal features and learn to predict eventual failure [1]. Others score how unusual those features are relative to successful rollouts [2]–[4].

These detectors are usually evaluated on completed rollouts, where a strong shortcut hides. Successful rollouts stop at first success, while failures typically run to the timeout. In our data, completed-rollout duration alone reaches ROC-AUC 0.80 on a held-out test of 76 rollouts, against 0.53 for the raw SAFE score and 0.69 after per-task normalization. SAFE avoids this shortcut by truncating rollouts to equal length [1], but a deployed detector cannot truncate. A causal version of this shortcut is legitimate at deployment: the probability that the task will fail, given that it is still running at step t. Any learned detector must therefore beat this elapsed-time risk at the same false-positive budget, not just chance.

We make this comparison under a strict protocol. Detectors are developed on one cohort, their thresholds are frozen on a separate calibration cohort at a 5% event-level FPR, and they are then evaluated once on a disjoint prospective cohort. We write down how each outcome will be interpreted before opening each new result. Our contributions are:

1) Elapsed time as the reference. A task-conditioned time-risk detector catches 63.2% of failures at 4.7% FPR on 760 prospective rollouts of Xiaomi-Robotics-1 on 38 RoboCasa365 tasks. Policy-feature scores used alone do no better at the same budget.
2) Time-aware fusion. Alarming when either a policy-internal score or the time risk crosses a jointly calibrated threshold raises recall to 95.5% at 5.8% FPR. [PENDING: confirmatory result on a fresh cohort.]
3) The fusion matters more than the representation. SAFE, Mahalanobis, kNN, PCA-KMeans and RND scores all reach 90–95% recall once fused with time. Their differences from fused SAFE are not significant, although the data cannot exclude differences of up to 0.15 recall. Random noise or label-free resampled SAFE scores fused the same way add at most about 3 points over time alone, so the gain needs a policy-internal signal, not just a score that grows with rollout length.

Our original, preregistered design restricted the policy score to the first quarter of the rollout and gained only 3.7 points of recall over time alone; we report it as such.

II. Related Work

Failure detection from policy internals. SAFE trains a small MLP or LSTM on a VLA's internal features to predict eventual failure, and thresholds it with functional conformal prediction [1]; we reuse its implementation. SAFE notes that a detector could simply count elapsed time and removes this shortcut by truncating rollouts to equal length. We instead keep elapsed time, as a calibrated, deployable baseline and as a signal to fuse. In machine-learning terms, duration is a shortcut feature [5], but at deployment its causal form is legitimate evidence. Recent probes improve labels, robustness or calibration [6]–[8], and VLA-Scope adds execution-progress features to a learned predictor [9]. None compares against elapsed time alone.

Detectors without failure data. FAIL-Detect scores policy inputs and outputs against successful rollouts only [2]. FIPER alarms when an RND score on policy embeddings and an action-chunk entropy both exceed conformal thresholds [10]. Requiring both signals suits FIPER's aim of suppressing false alarms. Our fusion instead fires when either signal crosses, so that each can catch failures the other misses, and the joint calibration keeps the combined FPR within budget. On our data, AND-fusion calibrated the same way reached higher test recall (1.000) but overshot the FPR budget further (0.089, Wilson 95% upper bound 0.122).

Consistency and VLM monitors. STAC [11], VLA-FAIL [12] and ActProbe [13] compare the overlapping parts of consecutive action chunks. Xiaomi-Robotics-1 executes each 16-step chunk in full, so this signal is undefined for our policy. Language and vision-language monitors judge task progress, detect anomalies or explain failures from frames [11], [14]–[17] and are complementary to the time signal.

Time-to-event modelling. Our time risk is a per-task survival-style estimate of the probability of failure given that a rollout is still running, in the tradition of Kaplan–Meier and hazard models [18], [19]. Deciding when to stop observing a sequence and declare its class is studied as early time-series classification [20]; our recall-versus-horizon analysis is its accuracy-earliness trade-off.

OOD scores and calibration. We use SAFE's embedding baselines: Mahalanobis distance [3], k-nearest-neighbour distance [4], PCA plus k-means distance, and random network distillation [21], all fitted on successful rollouts. Conformal thresholds give finite-sample false-positive control under exchangeability [22]. Most prior evaluations report ROC-AUC and one operating point on the test pool. We instead freeze every threshold on a separate calibration cohort and compare detectors at a matched event-level FPR on a disjoint prospective cohort.

III. Problem Setting and Metrics

A detector watches one rollout of a known task k and decides online, at each policy inference $t = 1, 2, \ldots$ (one inference per 16-step action chunk), whether to raise an alarm. The task horizon H_k is the length, in inferences, of the longest failed development rollout of task k, a proxy for the task timeout. Its score at t may use only what is available by t: the policy features so far, the inference index and the task identity. It may not use the rollout's final length, future features, or subtask annotations. The first threshold crossing is the alarm; the detector never changes the policy's actions.

Failure is the positive class. A failed rollout is detected if any alarm fires before it ends; a successful rollout with any alarm is a false positive. We report event-level TPR and FPR, balanced accuracy (BA), the miss-adjusted detection fraction—the first-alarm time as a fraction of the task horizon H_k, averaged over all failures with a missed failure counted as 1.0 (lower is earlier)—and recall at 10%, 25% and 50% of H_k. All detectors are compared at operating points fixed on the calibration cohort under FPR $\leq$ 0.05. Confidence intervals come from a paired bootstrap that resamples tasks, then rollouts within each task (2,000 replicates).

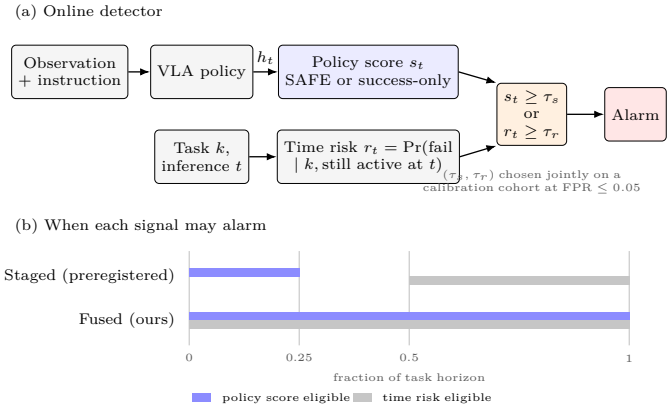


Fig. 1. Time-aware fusion. (a) At each policy inference, a policy-internal score and a task-conditioned elapsed-time risk are compared with two thresholds; the first crossing of either raises the alarm. Both thresholds are set together on a separate calibration cohort. (b) The preregistered staged design let the policy score alarm only in the first quarter of the rollout and time only after half; fusion lets both alarm throughout.

IV. Method

Time-aware fusion (Fig. 1) combines two causal scores, each with its own threshold, and alarms as soon as either crosses. Both thresholds are chosen together on a calibration cohort so that the combined detector meets the FPR budget.

A. Policy-internal score

SAFE. At each inference we take the final-layer action-token hidden state of the policy's flow-matching DiT, just before its action output layer (1,024 dimensions, last flow step, last action position). The official SAFE MLP maps it to a failure score $u_t^{(j)}$ for each of three training seeds j. With per-task location $m_k^{(j)}$ and scale $\sigma_k^{(j)}$ (median, and the larger of the scaled MAD and the standard deviation, of each development rollout's maximum score), the policy score is

$$s_t = \frac{1}{3}\sum_{j=1}^{3} \frac{\max_{t' \leq t} u_{t'}^{(j)} - m_k^{(j)}}{\sigma_k^{(j)}}. \qquad (1)$$

The running maximum does not change when a threshold is first crossed; it makes s_t monotone. We use the MLP head because it ranked failures best after task normalization on held-out development data (ROC-AUC 0.69 against 0.65 for the LSTM).

Success-only scores. Mahalanobis, cosine kNN, PCA-KMeans and feature-space RND are fitted on successful

development rollouts only, so they need no failure labels. Each produces a per-inference distance, aggregated causally (development selected a cumulative sum for all four) and normalized per task. Hyperparameters were chosen on held-out development data by early at-risk ROC-AUC.

B. Elapsed-time risk

The time risk is the probability that a still-running rollout of task k will fail:

$$r_t = \Pr(\text{failure} \mid \text{task } k, \text{ rollout still active at inference } t). \tag{2}$$

Among the development rollouts of task k still active at t, let n_t be their number and f_t the number that fail. We take the smoothed ratio $(f_t+0.5)/(n_t+1)$ for $t = 1, \ldots, H_k$ and fit an increasing isotonic regression in t/H_k, weighted by n_t. This is a survival-style estimate with no policy features. It is causal, because it uses only whether the rollout is still running. It needs failed development rollouts, but only to estimate this curve.

C. Calibrated fusion

The fused detector alarms at the first inference where either score crosses its threshold:

$$T_{\text{alarm}} = \min\{\, t : s_t \geq \tau_s \text{ or } r_t \geq \tau_r \,\}. \tag{3}$$

The pair (τ_s, τ_r) is searched jointly on the calibration cohort over every pair of per-rollout maxima observed on the 228 calibration rollouts. Among pairs with event-level FPR ≤ 0.05, we keep the one with the highest recall, then the earliest miss-adjusted detection, then the more conservative thresholds. A successful rollout counts as a false positive if either score alarms at any time.

D. Staged variant (preregistered)

Our original design restricted the policy score to the start of the rollout, where elapsed time cannot separate outcomes, and used time as a late fallback: the policy score may alarm only while $t \leq 0.25H_k$, and the time risk only from $t \geq 0.5H_k$. Thresholds are searched the same way. We froze this variant before the prospective test and report it as preregistered.

V. Experimental Setup

Policy and tasks. We use the released Xiaomi-Robotics-1 RoboCasa365 checkpoint [23] at a pinned revision. It predicts 16-step action chunks with a five-step flow-matching DiT and executes each chunk in full. We take the 39 official RoboCasa365 tasks [24], [25] whose published success count lies between 5 and 45 out of 50, and drop one with too few failures, leaving 38 tasks (10 atomic, 28 composite).

Cohorts. We use four disjoint cohorts, each for one purpose (Table II). Each evaluation cohort is collected from fresh environment seeds until every task has a fixed quota of successes and failures. The primary cohort takes the first rollouts of each class per task in seed order. Rollouts past the quota form a secondary all-retained cohort. No two cohorts share a seed or reset.

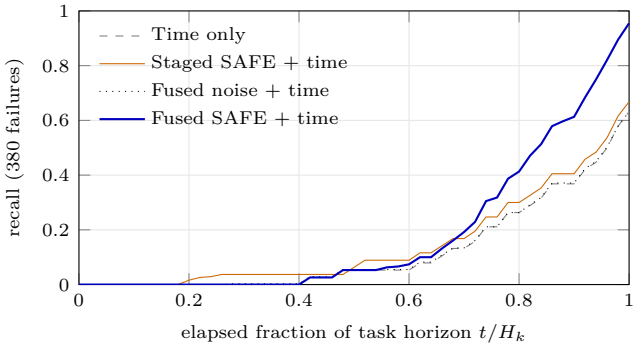


Fig. 2. Cumulative recall on the prospective test as a function of the elapsed fraction of the task horizon, at thresholds frozen on calibration (fused noise: one of 50 draws). Fusion adds detections late in the rollout; noise fused with time tracks time only.

Frozen protocol. The staged detector, all SAFE checkpoints, normalization and time curves were frozen before the prospective test was collected. The baselines and fused detectors were added later and evaluated once on the same opened cohort, with thresholds taken from calibration only. Each run first had to reproduce the frozen results and thresholds exactly and verify the checksums of the frozen inputs. Before each new evaluation we wrote down how we would interpret every outcome, including the rule for crediting SAFE specifically. These plans, and the confirmatory preregistration, are timestamped commits in our repository, to be released with the paper.

Detectors compared. Time only; SAFE only; staged SAFE + time (preregistered); fused SAFE + time; and each success-only score alone, staged with time, and fused with time. Two null scores test whether fusion only exploits longer failed rollouts, which give any score more chances to cross. Noise is the running maximum of i.i.d. Gaussian draws, one per inference. Resampled SAFE draws each inference's raw SAFE scores i.i.d. from all inferences of the same task in the same cohort, ignoring outcomes, and passes them through the frozen SAFE normalization. Each null is fused with time and calibrated exactly like fused SAFE, over 50 random draws. Finally, AND-fused SAFE + time alarms once both scores have crossed, with its own jointly calibrated thresholds. These controls were run after the review of this draft, on the same opened cohorts.

VI. Results

On the 760-rollout prospective test, every policy-internal score fused with elapsed time detects 90–96% of failures at about 6% FPR, against 63% for time alone (Table I, Fig. 2).

TABLE I

Prospective test (380 successes, 380 failures), thresholds frozen on calibration at FPR ≤ 0.05. BA = balanced accuracy; adj = miss-adjusted detection fraction (lower is earlier). Δ = paired recall difference with task-then-rollout bootstrap 95% CI. Fused rows were defined after this cohort was opened for the staged design. *One of 50 random draws; mean recall over draws 0.635 (noise) and 0.650 (resampled).

Detector	Recall	FPR	BA	adj	Δ recall vs. time	Δ recall vs. fused SAFE
Time only	0.632	0.047	0.792	0.880	–	−0.324 [−0.469, −0.178]
SAFE only	0.566	0.039	0.763	0.810	−0.065 [−0.354, +0.215]	−0.389 [−0.568, −0.218]
Success-only scores alone (4)	.582–.666	.055–.068	.763–.799	.782–.818	−0.050 to +0.035, CIs span 0	–
Staged SAFE + time (preregistered)	0.668	0.050	0.809	0.854	+0.037 [+0.000, +0.090]	−0.287 [−0.426, −0.156]
Staged kNN + time	0.666	0.061	0.803	0.857	+0.035 [+0.000, +0.097]	−0.289 [−0.431, −0.157]
Staged Mahal., PCA-KM, RND + time	0.639	.045–.050	.795–.797	.876–.877	+0.008 [+0.000, ≤+0.026]	–
Fused SAFE + time	0.955	0.058	0.949	0.816	+0.324 [+0.178, +0.469]	–
Fused kNN + time	0.947	0.068	0.939	0.769	+0.316 [+0.142, +0.495]	−0.008 [−0.093, +0.072]
Fused Mahalanobis + time	0.939	0.068	0.936	0.783	+0.308 [+0.133, +0.484]	−0.016 [−0.097, +0.065]
Fused PCA-KMeans + time	0.924	0.061	0.932	0.785	+0.293 [+0.105, +0.479]	−0.031 [−0.130, +0.061]
Fused RND + time	0.905	0.063	0.921	0.794	+0.275 [+0.082, +0.469]	−0.049 [−0.153, +0.050]
Fused noise + time (null)*	0.632	0.047	0.792	0.878	+0.000 [+0.000, +0.000]	−0.324 [−0.469, −0.178]
Fused resampled SAFE + time (null)*	0.655	0.063	0.796	0.869	+0.023 [+0.000, +0.081]	−0.301 [−0.446, −0.162]
AND-fused SAFE + time	1.000	0.089	0.955	0.798	+0.369 [+0.220, +0.522]	+0.045 [+0.005, +0.104]

TABLE II

Cohorts. S/F = successes/failures. The primary cohort takes the first q rollouts of each class per task; "retained" adds every rollout collected past the quota.

Cohort	Used to	Primary (S/F, q)	Retained
Development	fit SAFE, scores, time	1,824	4,511
Calibration	set thresholds	228 (114/114, 3)	451
Prospective test	evaluate once	760 (380/380, 10)	1,643
Confirmatory	preregistered test	760 (380/380, 10)	pending

TABLE III

Recall / FPR by task type on the prospective test (atomic: 10 tasks, 100 rollouts per class; composite: 28 tasks, 280 per class). Descriptive only: thresholds were set on the pooled calibration cohort, so per-stratum FPR is not controlled.

Detector	Atomic	Composite
Time only	0.400 / 0.020	0.714 / 0.057
SAFE only	0.800 / 0.080	0.482 / 0.025
Staged SAFE + time	0.540 / 0.040	0.714 / 0.054
Fused SAFE + time	0.980 / 0.030	0.946 / 0.068
Fused kNN + time	0.940 / 0.080	0.950 / 0.064
Fused Mahalanobis + time	0.890 / 0.050	0.957 / 0.075

A. Elapsed time is a strong reference

Time only reaches 0.632 recall at 0.047 FPR without any policy feature. SAFE alone reaches 0.566 at 0.039. The success-only scores alone reach 0.58–0.67, but at FPRs of 0.055–0.068, and their cumulative-sum aggregation grows with elapsed time, so part of their signal is duration.

B. Fusion beats staging

Fused SAFE + time detects 109 more failures than the staged design (363 against 254 of 380) for 3 more false alarms (22 against 19 of 380). Its miss-adjusted detection is 0.038 of the horizon lower (95% CI 0.006–0.070), but this metric counts each miss as 1.0 and so mostly reflects the extra detections. On the 240 failures that both fused SAFE and time only detect, the fused alarm is earlier for 8 and simultaneous for 232: fusion adds detections, it does not advance them. Fused SAFE flags a detected failure a median of 12 inferences (interquartile range 6–35), or 192 control steps, before the rollout ends; time only leaves 16.5 (5–52). Most of the policy score's complementary evidence arrives after the first quarter of the rollout, which the staged design discards.

C. The fusion matters more than the representation

No success-only score fused with time differs significantly from fused SAFE; the point gaps are 0.008–0.049 recall. The intervals are wide, however, and do not exclude differences of up to 0.15, so we claim no equivalence. Before evaluating, we had declared that SAFE would be credited only if every fused baseline's recall difference had a CI entirely below zero. None did, so we attribute the gain to fusing a policy-internal score with elapsed time. Fused SAFE has the lowest test FPR among fused detectors (0.058 against 0.061–0.068), a difference we did not test. Fused kNN alarms 0.047 of the horizon earlier than fused SAFE (95% CI 0.006–0.089) at a higher FPR. Inside the staged wrapper, the baselines trail SAFE by 0.002–0.029 recall, with every CI reaching zero. The choice of score also matters little for warning lead, the fraction of the time limit left at the first alarm, measured against a per-task timer that alarms past the 95th percentile of each task's development success durations: in an exploratory follow-up on the confirmatory cohorts, the four success-only scores fused with that timer gain 0.020–0.034 of the time limit over it, against 0.030 for SAFE.

Null controls. Fusion does not work with an uninformative score. Over 50 draws, noise fused with time reached 0.635 mean recall (range 0.632–0.650) at 0.054 mean FPR, and resampled SAFE reached 0.650 (0.637–0.663) at 0.053, both close to time only (Table I). On one

draw, fused SAFE beats fused resampled SAFE by 0.301 recall (95% CI 0.162–0.446). SAFE also carries length-matched evidence: among rollouts still running at 10, 25, 50 and 75% of the horizon, its within-task ROC-AUC is 0.55, 0.60, 0.59 and 0.58, against 0.51, 0.52, 0.53 and 0.50 for noise and exactly 0.5 for the time risk, which is constant within a task at a fixed step. The fused gain therefore needs a policy-internal signal, although which one matters little.

D. Atomic and composite tasks

Time alone is weak on atomic tasks and SAFE alone is weak on composites, and fusion repairs both (Table III). On atomic tasks fused SAFE has the best recall/FPR trade-off; on composites the fused scores are close. One hypothesis is a label mismatch: a terminal outcome label fits one atomic attempt, but marks the correct early work of a long composite task as failing. A first test did not support it: on five composite tasks, SAFE trained on segmented subtask labels ranked held-out stages no better than chance (ROC-AUC 0.49 against 0.65 for terminal labels). The gap is also not specific to SAFE (exploratory): on lead over the per-task timer, all five scores fused with it gain more on atomic (0.045–0.081) than on composite tasks (0.008–0.027), and an offline check on RLDX-1 shows the same split (SAFE's LSTM head: 0.049 against 0).

E. No early warning

No detector catches more than 2.9% of failures by 25% of the task horizon, and fused detectors catch none; staged SAFE + time reaches only 6.3% by half the horizon (Fig. 2). Fused SAFE + time catches 31% of failures by 75% of the horizon and 61% by 90%, against 21% and 37% for time only; the remaining third are caught in the last tenth. Fusion improves detection before the rollout ends, not early warning.

Most failures are never decided (exploratory). One labeler, blind to detector alarms, marked the point of no return (after which the policy could not finish in time without help) on 48 failures of the later confirmatory cohort (2 per atomic, 1 per composite task). 38 (79%) were still recoverable at the time limit (atomic 17/20, composite 21/28), most often slow or nearly successful; a second labeler agreed on this call for 11 of 12 videos (Cohen's $\kappa = 0.80$). On this benchmark, failure mostly means not finishing in time, which is why elapsed time is so informative, and a detector that reports failure early is mostly predicting slowness. All 10 decided failures were decided before 75% of the horizon.

The confound in a published benchmark (exploratory). In FIPER's released rollouts [10] (five tasks, two on a real robot), rollout length alone separates failures from successes with ROC-AUC 0.92–1.00 in four tasks (0.62 in the fifth). A timer calibrated like FIPER's conformal thresholds and scored with FIPER's own metrics reaches mean accuracy 0.83 against FIPER's published 0.78, but alarms at 0.63 of the horizon against 0.30, so FIPER wins on time-weighted accuracy (0.65 against 0.57): its advantage is earliness. These are published values against our computed timer, not a paired re-run; our horizon is the longest released rollout; and on the two real-robot tasks, with 10 calibration successes each, a finite-sample conformal threshold lowers the timer's accuracy to about 0.54. The boundary case is a fixed-horizon evaluation: in SAFE's real-robot release [1], 98% of failures and at least 30 successes per task stop at one fixed step, so elapsed time carries no outcome information there. Whether time is a strong baseline is decided by the termination rule.

F. Sensitivity to the test FPR

Fused detectors exceed the 0.05 budget on test (fused SAFE: 22 of 380, Wilson 95% interval 0.039–0.086), while time only stays at 0.047. The overshoot does not explain the gain. Even with its threshold chosen on the test cohort itself, time only reaches at most 0.711 recall at test FPR ≤ 0.058 and 0.737 at ≤ 0.068, against 0.955 for fused SAFE + time.

G. Confirmatory test

[PENDING: the preregistered test of fused SAFE + time against time only on 760 fresh rollouts (base seed 1300007), with fused kNN + time as the declared secondary. Criteria: bootstrap CI of the recall gain above zero, a point gain of at least 0.15, and test FPR at most 0.075 with Wilson upper bound at most 0.10.]

VII. Discussion and Limitations

The practical lesson is to evaluate any learned failure detector against a calibrated elapsed-time detector, and to deploy it fused with one. We hypothesize that the two signals catch different failures: time catches failures in tasks whose successes finish quickly, while the policy score catches failures in tasks where successful and failed rollouts run equally long. The data only partly support this. At the fused thresholds, 210 of the 380 failures cross only the time threshold, 123 only the policy-score threshold, 30 both and 17 neither, so the two signals do catch mostly different failures. But the policy-only catches are not concentrated in tasks whose success and failure durations overlap (Spearman $\rho = 0.12$, $p = 0.49$, against per-task duration ROC-AUC). Because success-only scores work about as well as SAFE once fused, failed rollouts are needed only to estimate the time-risk curve, not to train the policy score.

Limitations. (i) The fused detectors were defined after the prospective cohort had been opened for the staged design; their thresholds came only from calibration and each was evaluated once, but only the confirmatory test makes the headline claim preregistered. (ii) Every detector held 0.044 FPR on calibration, while fused detectors reached 0.058–0.068 on the test, with Wilson intervals that include 0.05; calibration had only 114 successful

rollouts. AND-fusion overshoots more (0.089). The null controls and AND-fusion were added after the cohort was opened. (iii) Alarms come late in the rollout, which may leave little time to recover; whether an alarm improves task success through reset-and-retry is untested. (iv) All results are in simulation and use one policy prospectively; a second (RLDX-1) was checked only offline, on existing data with a small test set. Earlier experiments with π_0 and RLDX-1 showed the same atomic-versus-composite gap on smaller datasets. (v) Task normalization and the time curves assume the deployed task and its horizon are known. (vi) Test cohorts are balanced by quota, so recall and FPR are reported per class rather than at deployment prevalence.

VIII. Conclusion

Elapsed time is the baseline a VLA failure detector must beat, and fusing it with a policy-internal score is the simplest way to beat it. On 760 prospective rollouts across 38 RoboCasa365 tasks, calibrated fusion detected 95.5% of failures at 5.8% FPR, against 63.2% for time alone, with no significant difference between the policy scores fused. Stage-aware supervision that gives early, localized warnings on long-horizon tasks is the natural next step. [PENDING: confirmatory result.]

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